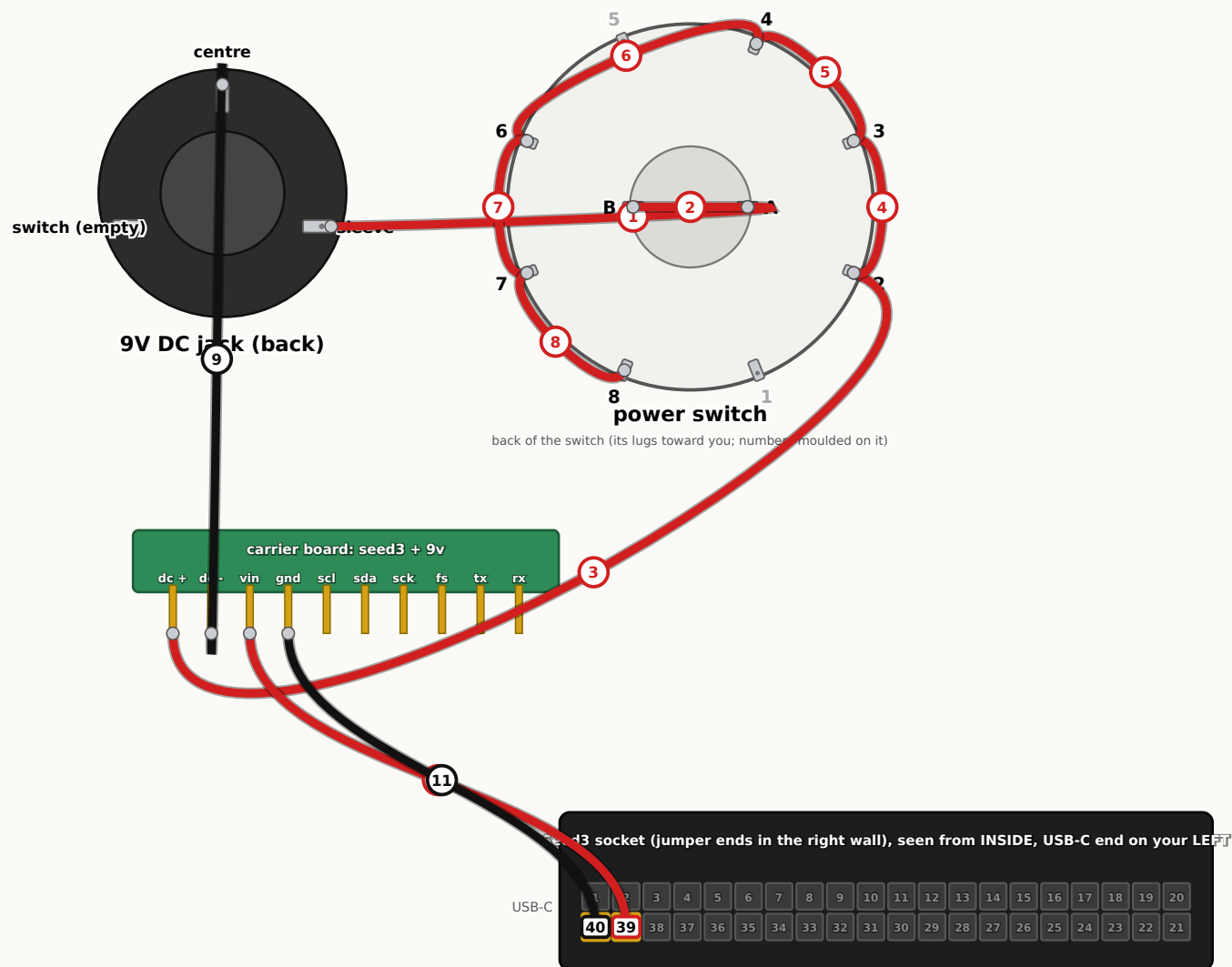


PG-1 wiring, page 1 of 10: Power: 9V jack -> power switch -> carrier board -> Seed3

Black = -, red = +. The power switch: + goes into BOTH commons (A and B); lugs 2 3 4 and 6 7 8 are all ON, so join them with bare offcuts and take ONE wire from them to 'dc +'. Lugs 1 and 5 = OFF: leave them empty.

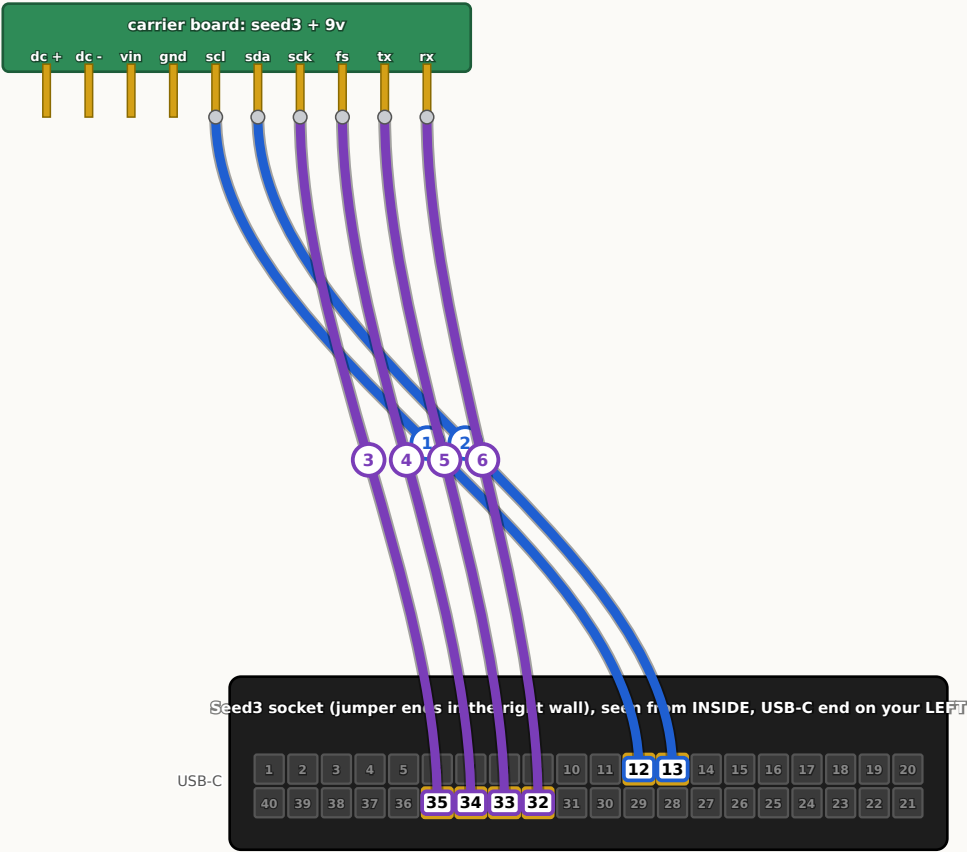


Steps (the number is on the wire):

- 1 9V jack: the SLEEVE lug (+) -> power switch: common A (the inner lug marked A) (9 V + into the switch (solder))
- 2 power switch: common A (the inner lug marked A) -> power switch: common B (the inner lug marked B) (join both commons (short bare offcut))
- 3 power switch: lug 2 (pole A, position 2) -> board 'seed3 + 9v' header: pin 'dc +' (1st from the left) (ON lugs -> dc + (join A2-A4 and B2-B4 with bare offcuts first))
- 4 power switch: lug 2 (pole A, position 2) -> power switch: lug 3 (pole A, position 3) (join (bare offcut))
- 5 power switch: lug 3 (pole A, position 3) -> power switch: lug 4 (pole A, position 4) (join (bare offcut))
- 6 power switch: lug 4 (pole A, position 4) -> power switch: lug 6 (pole B, position 2) (join (bare offcut))
- 7 power switch: lug 6 (pole B, position 2) -> power switch: lug 7 (pole B, position 3) (join (bare offcut))
- 8 power switch: lug 7 (pole B, position 3) -> power switch: lug 8 (pole B, position 4) (join (bare offcut))
- 9 9V jack: the CENTRE-PIN lug (its - : centre negative) -> board 'seed3 + 9v' header: pin 'dc -' (2nd from the left) (9 V - straight to the board (female end on the header))
- 10 board 'seed3 + 9v' header: pin 'vin' (3rd from the left) -> Seed3 socket place 39 (bottom row) (jumper: Seed3 power (9 V, protected))
- 11 board 'seed3 + 9v' header: pin 'gnd' (4th from the left) -> Seed3 socket place 40 (bottom row) (jumper: Seed3 ground (the ground chain is spliced onto this wire))

PG-1 wiring, page 2 of 10: Carrier board -> Seed3: the 6 signal jumpers

Plug the jumpers onto the board's header BEFORE the board goes in the box (the names are printed on the board's face side). The other ends go in the Seed3 socket places shown.

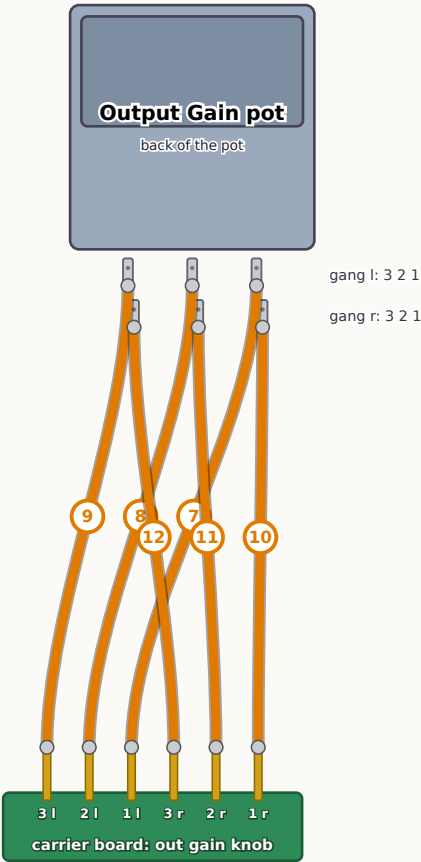
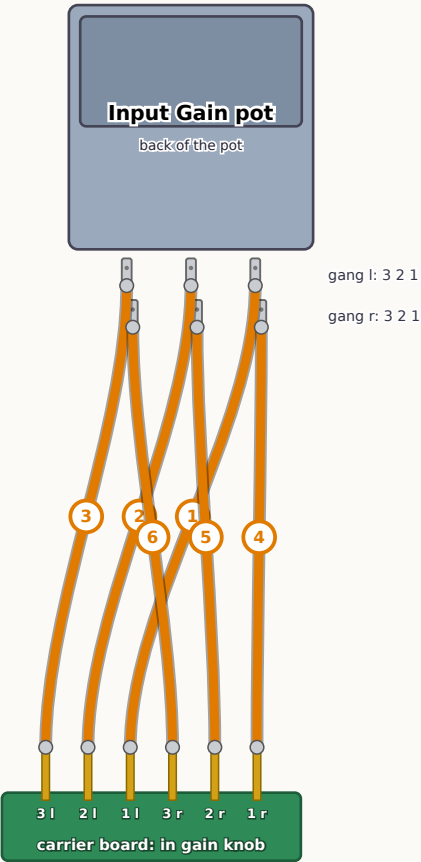


Steps (the number is on the wire):

- 1 board 'seed3 + 9v' header: pin 'scl' (5th from the left)
-> Seed3 socket place 12 (top row) (jumper: I2C clock)
- 2 board 'seed3 + 9v' header: pin 'sda' (6th from the left)
-> Seed3 socket place 13 (top row) (jumper: I2C data)
- 3 board 'seed3 + 9v' header: pin 'sck' (7th from the left)
-> Seed3 socket place 35 (bottom row) (jumper: audio bit clock)
- 4 board 'seed3 + 9v' header: pin 'fs' (8th from the left)
-> Seed3 socket place 34 (bottom row) (jumper: audio frame clock)
- 5 board 'seed3 + 9v' header: pin 'tx' (9th from the left)
-> Seed3 socket place 33 (bottom row) (jumper: audio Seed3 -> board)
- 6 board 'seed3 + 9v' header: pin 'rx' (last on the right)
-> Seed3 socket place 32 (bottom row) (jumper: audio board -> Seed3)

PG-1 wiring, page 3 of 10: The gain pots -> their headers on the carrier board

Each pot has 2 rows of 3 pins: the row nearer the PANEL is gang 'l', the row nearer YOU is gang 'r'. Short wires soldered on the pot, female jumper ends on the header (header names are printed on the board).

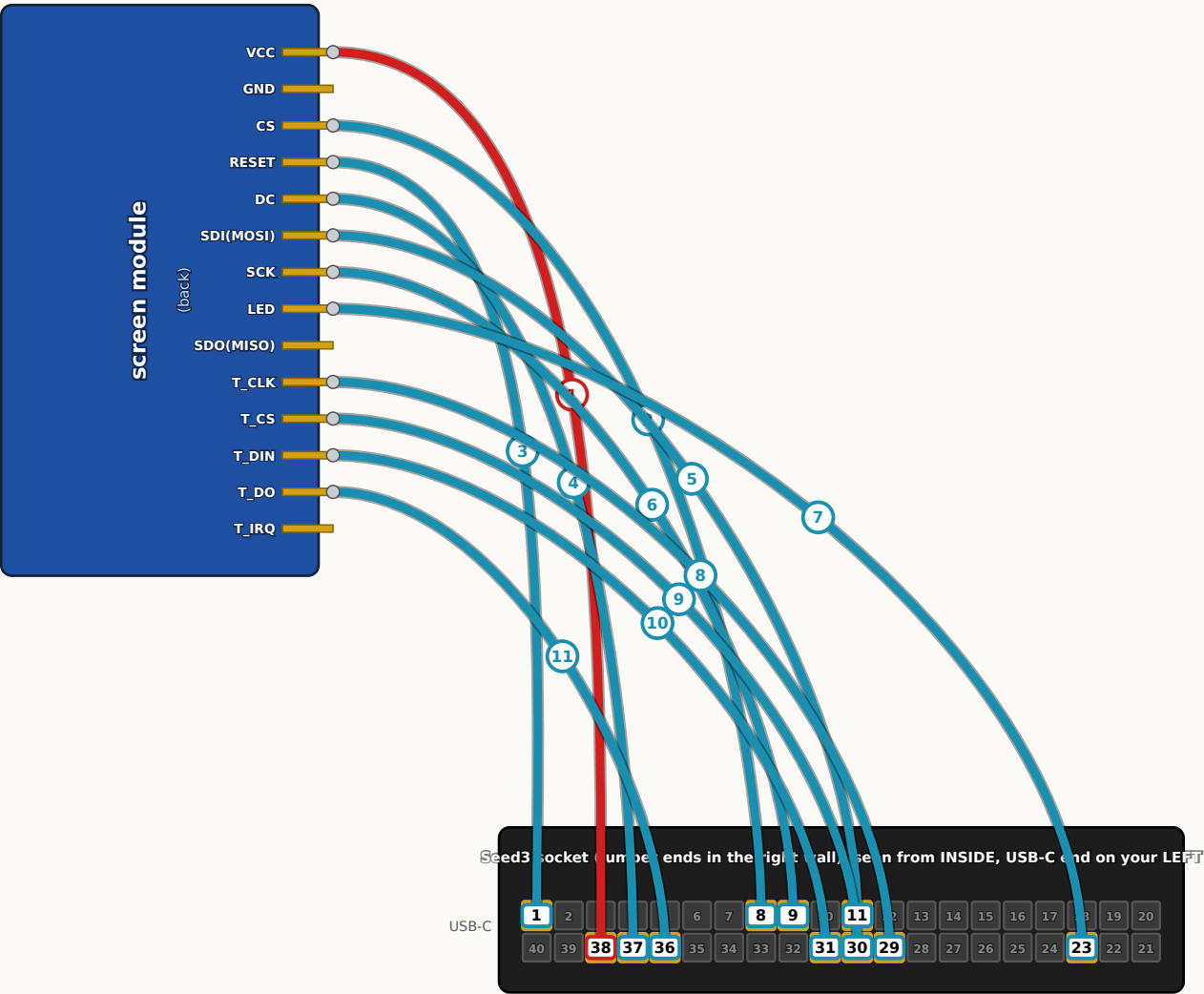


Steps (the number is on the wire):

- 1 Input Gain pot: pin 1 l (RIGHT pin of the row nearer the PANEL) -> board 'in gain knob' header: pin '1 l' (3rd from the left)
- 2 Input Gain pot: pin 2 l (MIDDLE pin of the row nearer the PANEL) -> board 'in gain knob' header: pin '2 l' (2nd from the left)
- 3 Input Gain pot: pin 3 l (LEFT pin of the row nearer the PANEL) -> board 'in gain knob' header: pin '3 l' (1st from the left)
- 4 Input Gain pot: pin 1 r (RIGHT pin of the row nearer YOU) -> board 'in gain knob' header: pin '1 r' (last on the right)
- 5 Input Gain pot: pin 2 r (MIDDLE pin of the row nearer YOU) -> board 'in gain knob' header: pin '2 r' (5th from the left)
- 6 Input Gain pot: pin 3 r (LEFT pin of the row nearer YOU) -> board 'in gain knob' header: pin '3 r' (4th from the left)
- 7 Output Gain pot: pin 1 l (RIGHT pin of the row nearer the PANEL) -> board 'out gain knob' header: pin '1 l' (3rd from the left)
- 8 Output Gain pot: pin 2 l (MIDDLE pin of the row nearer the PANEL) -> board 'out gain knob' header: pin '2 l' (2nd from the left)
- 9 Output Gain pot: pin 3 l (LEFT pin of the row nearer the PANEL) -> board 'out gain knob' header: pin '3 l' (1st from the left)
- 10 Output Gain pot: pin 1 r (RIGHT pin of the row nearer YOU) -> board 'out gain knob' header: pin '1 r' (last on the right)
- 11 Output Gain pot: pin 2 r (MIDDLE pin of the row nearer YOU) -> board 'out gain knob' header: pin '2 r' (5th from the left)
- 12 Output Gain pot: pin 3 r (LEFT pin of the row nearer YOU) -> board 'out gain knob' header: pin '3 r' (4th from the left)

PG-1 wiring, page 4 of 10: Screen -> Seed3

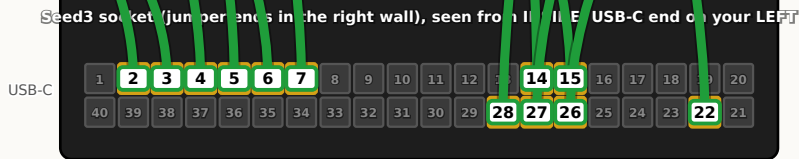
Read the names printed beside the screen's pins. 'SDO(MISO)' and 'T_IRQ' stay empty. 'GND' is on the ground-chain page.



Steps (the number is on the wire):

- 1 screen: pin 1 'VCC' (printed on the module) -> Seed3 socket place 38 (bottom row) (jumper: 3.3 V)
- 2 screen: pin 3 'CS' (printed on the module) -> Seed3 socket place 8 (top row)
- 3 screen: pin 4 'RESET' (printed on the module) -> Seed3 socket place 1 (top row)
- 4 screen: pin 5 'DC' (printed on the module) -> Seed3 socket place 37 (bottom row)
- 5 screen: pin 6 'SDI(MOSI)' (printed on the module) -> Seed3 socket place 11 (top row)
- 6 screen: pin 7 'SCK' (printed on the module) -> Seed3 socket place 9 (top row)
- 7 screen: pin 8 'LED' (printed on the module) -> Seed3 socket place 23 (bottom row) (jumper: backlight)
- 8 screen: pin 10 'T_CLK' (printed on the module) -> Seed3 socket place 29 (bottom row) (jumper: touch)
- 9 screen: pin 11 'T_CS' (printed on the module) -> Seed3 socket place 30 (bottom row) (jumper: touch)
- 10 screen: pin 12 'T_DIN' (printed on the module) -> Seed3 socket place 31 (bottom row) (jumper: touch)
- 11 screen: pin 13 'T_DO' (printed on the module) -> Seed3 socket place 36 (bottom row) (jumper: touch)

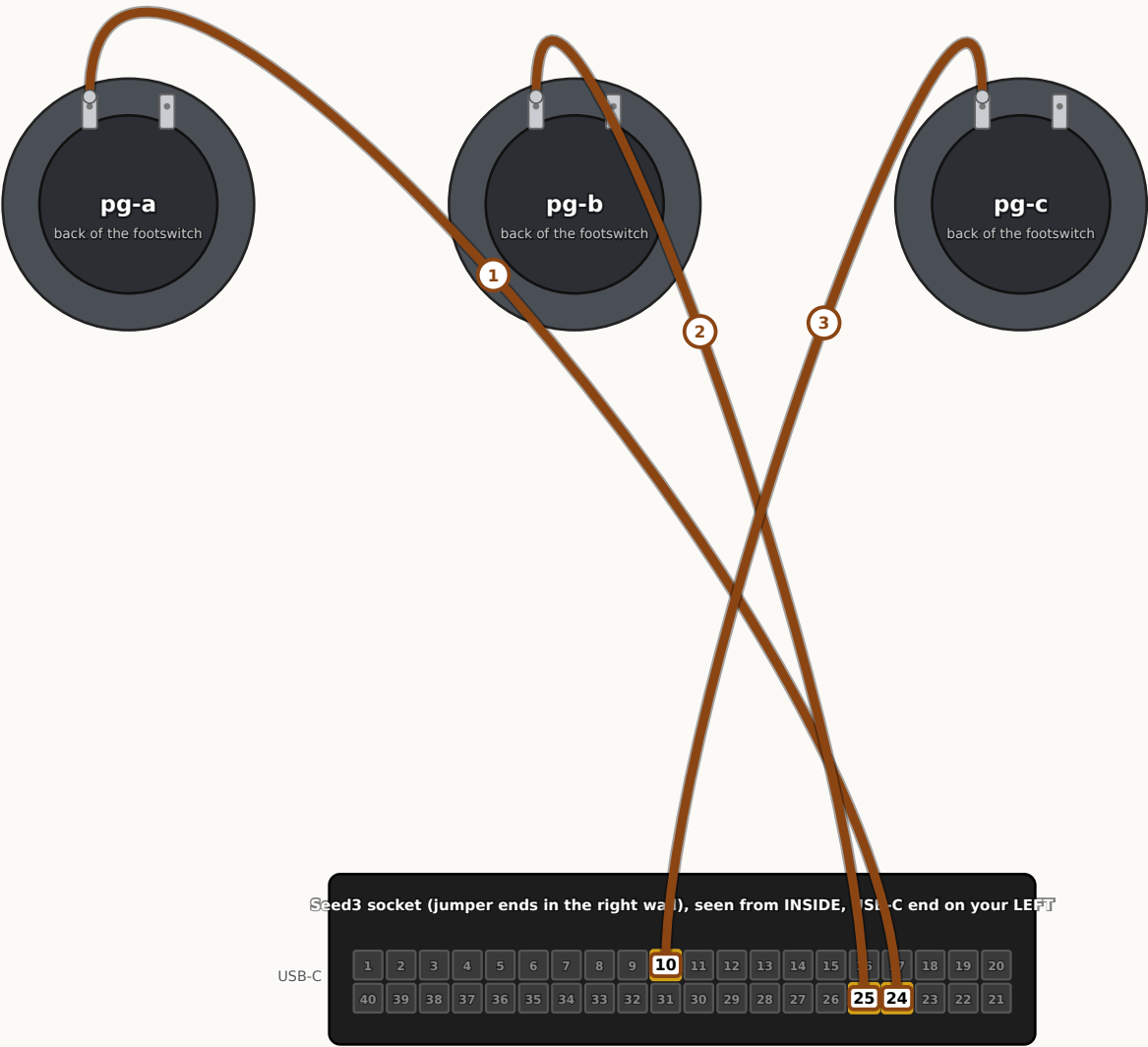
Back of each encoder: the row of 3 pins at the bottom, the 2 push pins at the top. C and 'push 2' are on the ground-chain page. If a knob turns the wrong way, flip 'knobs' in the config tab (no rewiring).



- 1 pg-1: A (RIGHT pin of the row of 3) -> Seed3 socket place 2 (top row)
- 2 pg-1: B (LEFT pin of the row of 3) -> Seed3 socket place 3 (top row)
- 3 pg-1: push (LEFT pin of the 2) -> Seed3 socket place 4 (top row)
- 4 pg-2: A (RIGHT pin of the row of 3) -> Seed3 socket place 5 (top row)
- 5 pg-2: B (LEFT pin of the row of 3) -> Seed3 socket place 6 (top row)
- 6 pg-2: push (LEFT pin of the 2) -> Seed3 socket place 7 (top row)
- 7 pg-3: A (RIGHT pin of the row of 3) -> Seed3 socket place 14 (top row)
- 8 pg-3: B (LEFT pin of the row of 3) -> Seed3 socket place 15 (top row)
- 9 pg-3: push (LEFT pin of the 2) -> Seed3 socket place 22 (bottom row)
- 10 pg-4: A (RIGHT pin of the row of 3) -> Seed3 socket place 26 (bottom row)
- 11 pg-4: B (LEFT pin of the row of 3) -> Seed3 socket place 27 (bottom row)
- 12 pg-4: push (LEFT pin of the 2) -> Seed3 socket place 28 (bottom row)

PG-1 wiring, page 6 of 10: The 3 footswitches -> Seed3

Either lug of each footswitch to its Seed3 place; the other lug is on the ground-chain page.

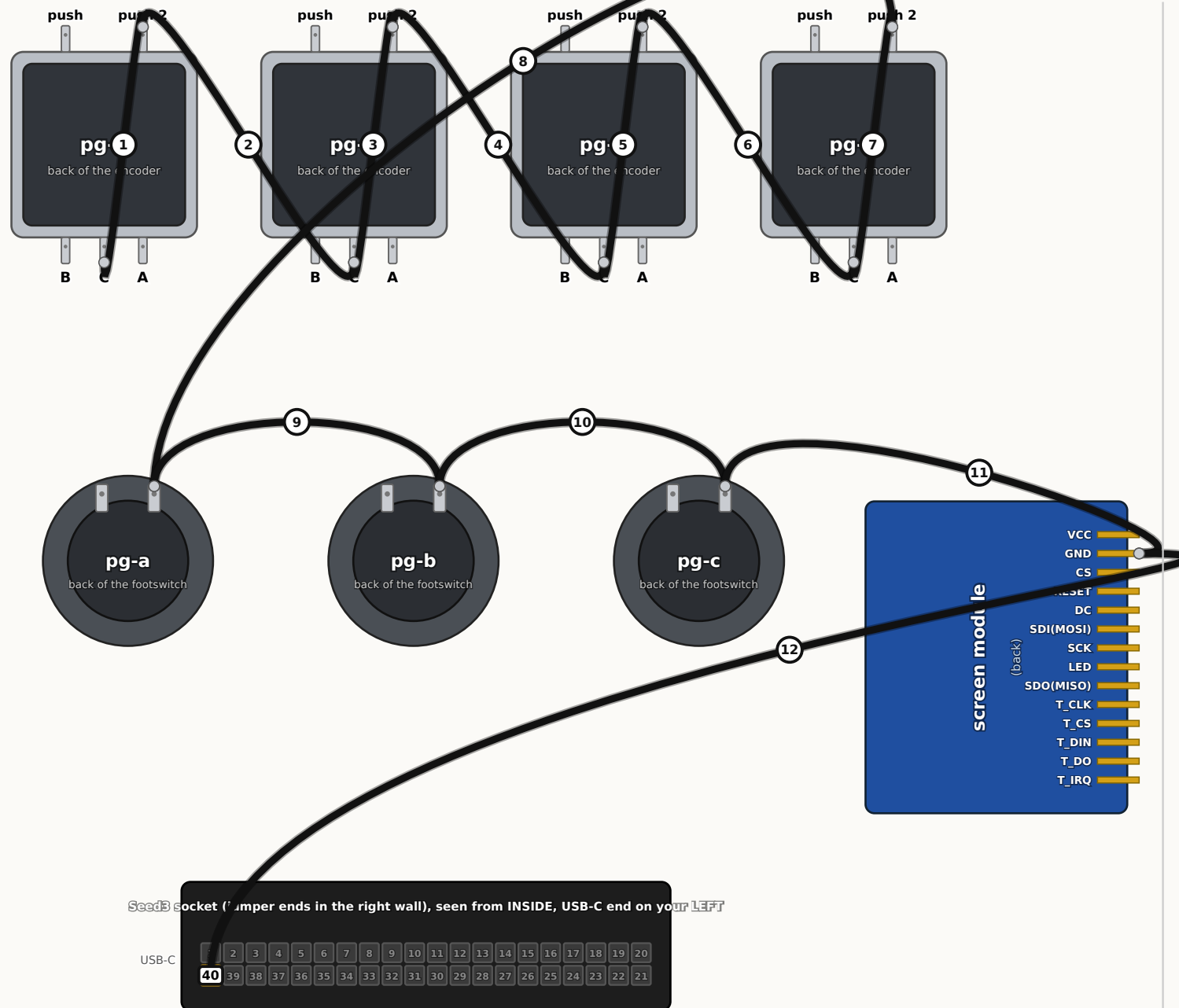


Steps (the number is on the wire):

- 1 pg-a: either lug -> Seed3 socket place 24 (bottom row)
- 2 pg-b: either lug -> Seed3 socket place 25 (bottom row)
- 3 pg-c: either lug -> Seed3 socket place 10 (top row)

PG-1 wiring, page 7 of 10: The ground chain (one black wire, part to part)

Solder a short black wire from each lug to the next, in the order of the steps. The last one (from the screen's GND) is **SPLICED** onto the jumper from the board's 'gnd' to Seed3 place 40, about 3 cm from the socket: strip 5 mm of that jumper, wrap, solder, heat shrink (a socket place holds only one jumper end).

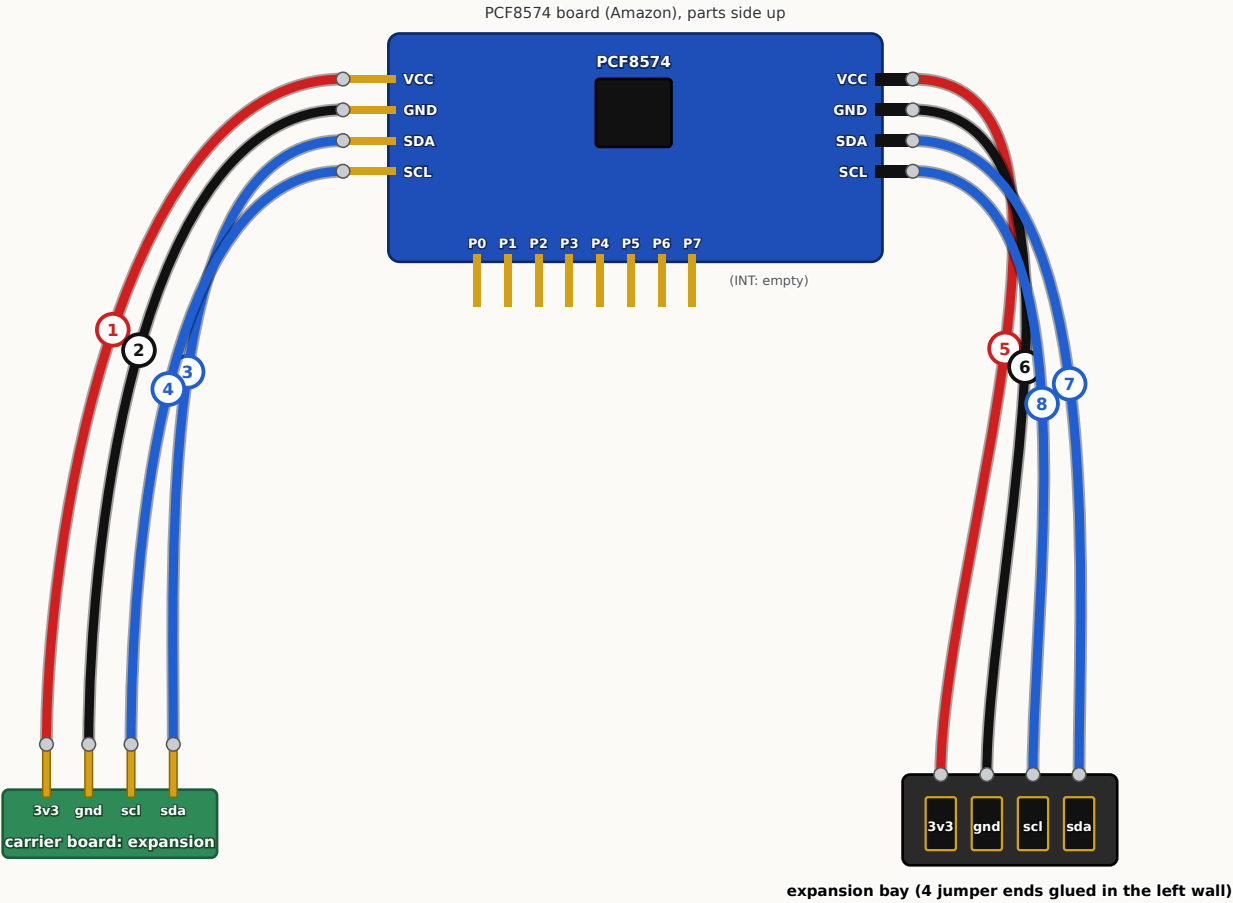


Steps (the number is on the wire):

- 1 pg-1: C (MIDDLE pin of the row of 3) -> pg-1: push 2 (RIGHT pin of the 2)
- 2 pg-1: push 2 (RIGHT pin of the 2) -> pg-2: C (MIDDLE pin of the row of 3)
- 3 pg-2: C (MIDDLE pin of the row of 3) -> pg-2: push 2 (RIGHT pin of the 2)
- 4 pg-2: push 2 (RIGHT pin of the 2) -> pg-3: C (MIDDLE pin of the row of 3)
- 5 pg-3: C (MIDDLE pin of the row of 3) -> pg-3: push 2 (RIGHT pin of the 2)
- 6 pg-3: push 2 (RIGHT pin of the 2) -> pg-4: C (MIDDLE pin of the row of 3)
- 7 pg-4: C (MIDDLE pin of the row of 3) -> pg-4: push 2 (RIGHT pin of the 2)
- 8 pg-4: push 2 (RIGHT pin of the 2) -> pg-a: the other lug
- 9 pg-a: the other lug -> pg-b: the other lug
- 10 pg-b: the other lug -> pg-c: the other lug
- 11 pg-c: the other lug -> screen: pin 2 'GND' (printed on the module) (ground chain (screen GND: cut a jumper, solder its cut end in the chain))
- 12 screen: pin 2 'GND' (printed on the module) -> Seed3 socket place 40 (bottom row) (chain's end: SPLICED onto the j10.gnd -> seed.40 jumper, 3 cm from the socket (strip 5 mm, wrap, solder, heat shrink). A socket place holds one jumper only)

PG-1 wiring, page 8 of 10: PCF8574 board -> carrier board and the expansion bay

The 4 input pins go to the board's 'expansion' header; its 4 pass-through pins go to the bay in the left wall. The PCF8574 board is stuck to the inside of the lid with velcro.

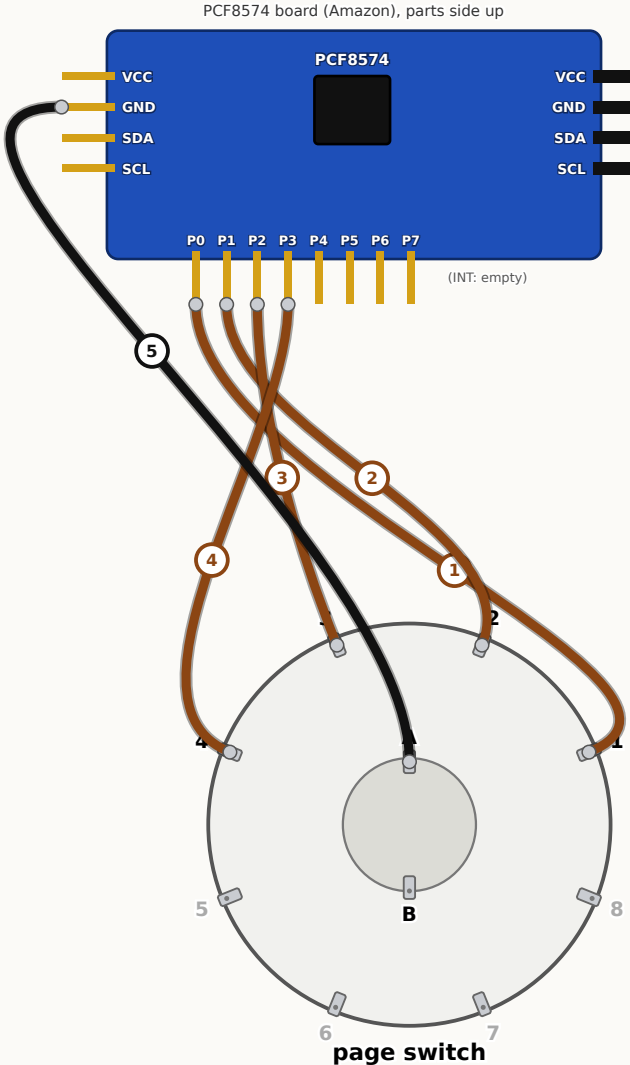


Steps (the number is on the wire):

- 1 PCF8574: 'VCC' on the INPUT pins (left end) -> board 'expansion' header: pin '3v3' (1st from the left) (jumper (3.3 V))
- 2 PCF8574: 'GND' on the INPUT pins (left end) -> board 'expansion' header: pin 'gnd' (2nd from the left)
- 3 PCF8574: 'SDA' on the INPUT pins (left end) -> board 'expansion' header: pin 'sda' (last on the right)
- 4 PCF8574: 'SCL' on the INPUT pins (left end) -> board 'expansion' header: pin 'scl' (3rd from the left)
- 5 PCF8574: 'VCC' on the PASS-THROUGH pins (right end) -> expansion bay: the '3v3' place (jumper, its far end glued in the bay)
- 6 PCF8574: 'GND' on the PASS-THROUGH pins (right end) -> expansion bay: the 'gnd' place (jumper, its far end glued in the bay)
- 7 PCF8574: 'SDA' on the PASS-THROUGH pins (right end) -> expansion bay: the 'sda' place (jumper, its far end glued in the bay)
- 8 PCF8574: 'SCL' on the PASS-THROUGH pins (right end) -> expansion bay: the 'scl' place (jumper, its far end glued in the bay)

PG-1 wiring, page 9 of 10: Page switch -> PCF8574 board

Pole A only (lugs 1 2 3 4 and common A); pole B (5 6 7 8, B) stays empty. Common A goes to GND: the GND pin already has the carrier's jumper on it, so solder this wire on the PCF board's UNDERSIDE, on GND's joint.



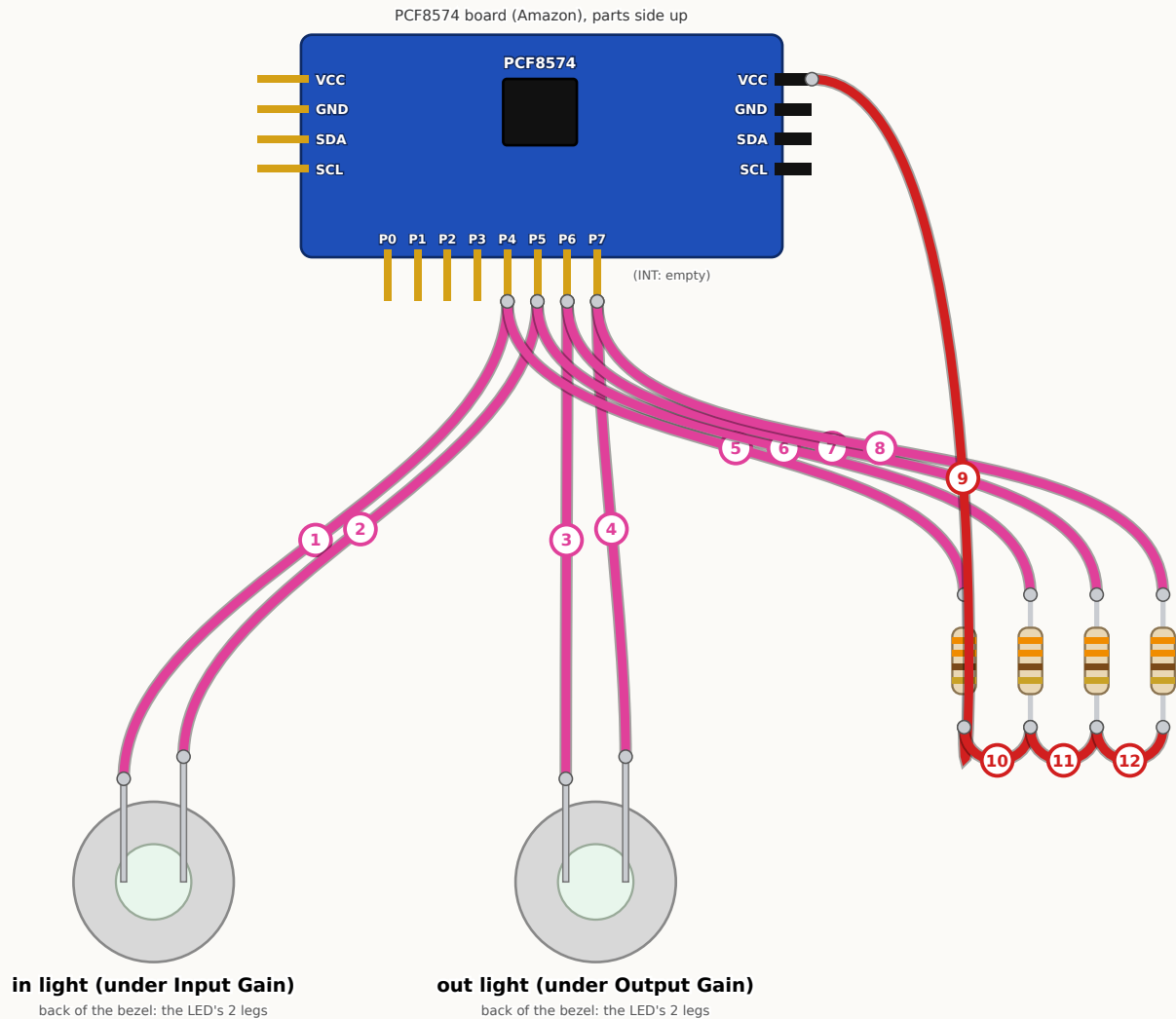
back of the switch (its lugs toward you; numbers moulded on it), drawn turned: it can sit at any angle

Steps (the number is on the wire):

- 1 page switch: lug 1 (pole A, position 1) -> PCF8574: 'P0' on the row of 8 (position 1 (find with beep mode))
- 2 page switch: lug 2 (pole A, position 2) -> PCF8574: 'P1' on the row of 8 (position 2)
- 3 page switch: lug 3 (pole A, position 3) -> PCF8574: 'P2' on the row of 8 (position 3)
- 4 page switch: lug 4 (pole A, position 4) -> PCF8574: 'P3' on the row of 8 (position 4)
- 5 page switch: common A (the inner lug marked A) -> PCF8574: 'GND' on the INPUT pins (left end) (pole A common: solder it on the UNDERSIDE of the PCF board, on the GND pin's solder joint)

PG-1 wiring, page 10 of 10: Level lights + 330 ohm resistors -> PCF8574 board

Each light's 2 legs go to 2 P pins (either way round: if a light shows red with signal, swap its 2 wires). Each P4-P7 also gets a 330 ohm to VCC, soldered on the PCF board's UNDERSIDE; the 4 resistors' other ends are joined and go to a VCC pin.



Steps (the number is on the wire):

- 1 in light (under Input Gain): the LEFT leg (either way round) -> PCF8574: 'P4' on the row of 8 (in light (swap its 2 legs, or set kSwapLightColours, if the colours are backwards))
- 2 in light (under Input Gain): the RIGHT leg (either way round) -> PCF8574: 'P5' on the row of 8 (in light)
- 3 out light (under Output Gain): the LEFT leg (either way round) -> PCF8574: 'P6' on the row of 8 (out light)
- 4 out light (under Output Gain): the RIGHT leg (either way round) -> PCF8574: 'P7' on the row of 8 (out light)
- 5 330 ohm resistor 1: the TOP leg -> PCF8574: 'P4' on the row of 8 (330 ohm pull-up: solder its leg on the PCF board's underside, on P4's joint)
- 6 330 ohm resistor 2: the TOP leg -> PCF8574: 'P5' on the row of 8 (330 ohm pull-up)
- 7 330 ohm resistor 3: the TOP leg -> PCF8574: 'P6' on the row of 8 (330 ohm pull-up)
- 8 330 ohm resistor 4: the TOP leg -> PCF8574: 'P7' on the row of 8 (330 ohm pull-up)
- 9 330 ohm resistor 1: the BOTTOM leg -> PCF8574: 'VCC' on the PASS-THROUGH pins (right end) (the 4 resistors' other ends joined -> underside of a VCC pin's joint)
- 10 330 ohm resistor 1: the BOTTOM leg -> 330 ohm resistor 2: the BOTTOM leg (join)
- 11 330 ohm resistor 2: the BOTTOM leg -> 330 ohm resistor 3: the BOTTOM leg (join)
- 12 330 ohm resistor 3: the BOTTOM leg -> 330 ohm resistor 4: the BOTTOM leg (join)